

Carlos Manuel Hernández Hernández

AI-Driven Software Engineer

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Professional Summary

4th-year Computer Science student at UCLV with hands-on experience building AI-powered applications and data-driven automation systems. Reduced manual processing by 95% at ETECSA and cut data entry time by 40% via NLP integration in a freelance project. Building an end-to-end ML portfolio covering the full lifecycle: from data engineering and deep learning to model deployment with MLOps.

Experience

Software Engineer (Freelance, Remote) — AI-Powered Financial Bot

November 2025

- Built a production-grade Telegram bot for automated accounting and inventory management across multiple business units.
- Integrated OpenAI API for natural language processing of financial transactions, reducing manual data entry time by 40%.
- Architected an async Python backend with SQLite, generating real-time PDF/Excel financial reports for auditing.
- Implemented role-based access control and multi-currency support for distributed cash flow management.

University Intern — Asset Sync Inventory Automation, ETECSA

September – October 2024 · Telecommunications Company of Cuba

- Led development of an asset synchronization system automating reconciliation between financial databases and operational inventories.
- Developed bulk update algorithms with Python and Django, reducing manual processing time by 95% and eliminating human errors.
- Designed an analytical engine with Pandas and NumPy for real-time anomaly detection and data quality metrics.
- Implemented CI/CD pipelines with Docker and GitHub Actions for reliable production deployments.

Projects

ML Pipeline for NYC Taxi Trip Duration Prediction

github.com/cmhh22/ml-bigdata

- End-to-end ML pipeline with advanced feature engineering, XGBoost/LightGBM modeling, SHAP interpretability analysis, and a Streamlit web app for predictions. Big Data project with full EDA.

Transformer Experiments: Emotion Classification & RAG System

github.com/cmhh22/transformer-experiments

- BERT vs Qwen emotion classification comparison and a RAG (Retrieval-Augmented Generation) system using LangChain and ChromaDB for document-based Q&A.

Minecraft LLM Agent

github.com/cmhh22/minecraft-llm-agent

- AI-powered Minecraft bot using LLMs (GPT-4) for autonomous building, resource gathering, and conversation. Explores agent capabilities with Mineflayer + GitHub Models API.

PyTorch Deep Learning Collection

github.com/cmhh22/pytorch-deep-learning

- Comprehensive collection of PyTorch implementations covering MLPs, CNNs, RNNs, LSTMs, and transfer learning with detailed documentation.

EDA Storytelling: Airbnb NYC Analysis

github.com/cmhh22/eda-storytelling

- Exploratory Data Analysis project with advanced visualizations on Airbnb NYC data, focused on storytelling and actionable insights.

Education

Bachelor's Degree in Computer Science

Universidad Central "Marta Abreu" de Las Villas (UCLV) · 2021 – Present (4th Year)

Certifications & Courses

- HackerRank — Python (Basic), SQL (Basic)
- Kaggle — Python, Pandas, Intro to Machine Learning, Intermediate Machine Learning
- HuggingFace NLP Course (in progress)

Technical Skills

Languages	Python, JavaScript, TypeScript, SQL, C++, HTML5, CSS3
AI / ML	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, HuggingFace Transformers (BERT, Qwen), LLMs, RAG, LangChain
Backend & API	Django, FastAPI, Node.js, RESTful APIs, WebSockets (Socket.io)
Data & DB	PySpark, Pandas, NumPy, Dask, PostgreSQL, MySQL, SQLite, ChromaDB
DevOps / MLOps	Docker, MLflow, GitHub Actions, CI/CD, Git
Tools	Streamlit, Plotly, Matplotlib, Postman

Languages

Spanish (Native) · English (Professional Working Proficiency)